

Why the discovery map is not Google Maps (and why we used d3-geo)

This note is for the client conversation: **why we could not deliver the homepage map on Google Maps**, and **why we chose d3-geo** instead.

The short answer: the brief asks for a **RosoTravel-owned recommendation surface**, not a Google Maps product. Google Maps cannot honour several **non-negotiable** items in the client requirements. d3-geo can.

What the brief actually asks for

From *Destination Discovery Map* — *Client Requirements*:

Requirement	Where it is written
Phase 1 is a custom flat world map in Natural Earth projection	Phase scope + Level 0 map treatment
The product is not a Google Maps–style wayfinding tool	“What this map is NOT”
Own vector geometry , RosoTravel styling	Non-negotiables — Basemap
No third-party map branding , no provider POI labels on the surface	Same
Warm land, pale water, hairline national borders only	Level 0 map treatment
Recommended countries tinted a half-step warmer (and re-tinted when a preference chip is tapped)	Level 0 + Preference row
No provider labels, no city names from the basemap, no roads	Level 0 — Suppressed
Level 1 switches to Mercator , fitted to the country; neighbours visible but unlabelled	Level 1
Level 2 is a quiet dotted field with rings and a km scale — not a street map	Level 2
Phase 2 is a 3D globe with rotation/tilt in the same visual language	Phase scope

That is a designed **recommendation canvas**. Pins are opinions. Geography is context. Google Maps is built to be a **navigation product** (roads, places, Google’s world, Google’s attribution).

Why Google Maps cannot match this

We did implement a Google Maps version ([/google](#)) so the gap is visible, not theoretical. These limits are in the **Maps JavaScript API**, not in our code.

1. Natural Earth projection — not available

Google Maps is **Web Mercator** only. There is no API to draw the world in Natural Earth (or any other cartographic projection).

The brief's Phase 1 deliverable is explicitly "*Custom flat world map (Natural Earth projection)*". That single line rules Google Maps out as the world surface.

2. "No third-party branding" vs Google's terms

Google requires **visible Google attribution** (logo and data credits) on every map. Hiding it breaks their terms of use.

The brief says: **own vector geometry, RosoTravel styling, no third-party map branding**. We cannot legally ship a "Google-free" Google Map.

3. The basemap is Google's, not ours

Even with Cloud map styling we still get:

- Google's land/water/road model
- Google's labels and POIs unless a custom Map ID is fully restyled in Cloud Console
- Roads and city names that the brief says must be **suppressed**

JSON map styles (`styles: [...]`) are **ignored** when a **Map ID** is set. Advanced Markers (the HTML pills) **require** a Map ID. So the two things we need — custom pins and custom basemap colours — fight each other on Google's vector maps.

We can overlay country polygons on top of Google. That is a **skin**, not "own vector geometry". Roads and labels can still show through. Preference **re-tint** of recommended countries is unreliable.

4. Level 2 is not a Google map

City view is specified as: light dotted field, centre dot, distance rings, kilometre scale, quiet geography. That is a **diagram**, not Google roadmap or satellite. Forcing Google here puts roads and POIs back on a surface the brief wants silent.

5. Photorealistic 3D is the wrong Phase 2

Google's 3D is **photorealistic tiles** (heavy, branded, not RosoTravel land tints). The brief's Phase 2 is a **globe with rotation/tilt** in the same recommendation language as the flat map — not Street View in orbit.

6. The brief says it is not a wayfinding tool

Google Maps is wayfinding: pan the world, search places, follow roads. Using it as the homepage centrepiece fights the product definition, even if we hide the UI chrome.

Why d3-geo

[d3-geo](#) is a small, standard library for **projecting geographic coordinates** (lat/lng → x/y). It does not draw Google's world. We draw **our** polygons.

Brief item	What d3-geo lets us do
Natural Earth (L0)	<code>geoNaturalEarth1</code>
Mercator fitted to country (L1)	<code>geoMercator</code> + fit to the country polygon
Globe rotate/tilt (Phase 2)	<code>geoOrthographic</code> + drag to rotate
Own geometry	Natural Earth country outlines (via world-atlas / TopoJSON) — no Google tiles
Warm land, pale water, hairline borders	SVG fills and strokes we control (<code>--hm-land</code> , <code>--hm-sea</code> , <code>--hm-coast</code>)
Recommended countries warmer; re-tint on preference	Fill by match tier; update on chip tap
No provider labels / roads / POIs	We never draw them
No Google logo	Nothing in the Google Maps terms applies
City dotted field + rings	SVG on a quiet field; true lat/lng for attractions
Frame all destinations	Fit the projection to catalog points so pins stay in the card

We still use the **same catalog, ranking, pins, and panel**. Only the **map surface** changed. Funnel behaviour did not depend on Google.

Country outlines come from **Natural Earth** (public cartographic data), not from Google. That matches “own vector geometry”.

What we kept Google Maps for

Google Maps is still in the prototype at `/google`.

Use it to show stakeholders:

- Pins, preference re-rank, and the four levels **can** sit on Google
- The **look** (Natural Earth, no branding, quiet city diagram) **cannot**

It is a comparison surface, not the homepage. Production homepage should follow the spec map (`/`).

Honest limits (either engine)

Choosing d3-geo does **not** finish the whole brief. These still need the real (likely Next.js) build:

- Server-rendered, crawlable destination list and real `/destinations/...` pages
- Attraction clusters on country view
- Poster / first paint, loading skeleton, map-error fallback
- Preference-specific review slices, working “Check dates” CTA

Those are **product/platform** items, not reasons to go back to Google Maps.

Recommended line for the client

Google Maps cannot legally or technically be the RosoTravel homepage map as specified: it is Web Mercator only, it requires Google branding, and its basemap is a wayfinding product. We used d3-geo so Phase 1 can be Natural Earth with our colours, hairline borders, and preference tints, and Phase 2 can be a globe in the same language. Google Maps remains available as a comparison at [/google](#). In production we keep this custom surface and put crawlable destination pages around it.

One-page comparison

	Google Maps JS API	d3-geo + Natural Earth (current /)
Natural Earth world	No	Yes
RosoTravel land/water/borders	Partial overlay only; Map ID vs styles conflict	Yes
No third-party branding	No (attribution required)	Yes
Re-tint recommended countries	Unreliable	Yes
Quiet city diagram (no roads)	Fights the product	Yes
Globe in the same visual system	Photorealistic Google 3D (different product)	Orthographic globe
API key / Google bill	Yes	No
Street-level wayfinding	Yes (not asked for)	No (by design)